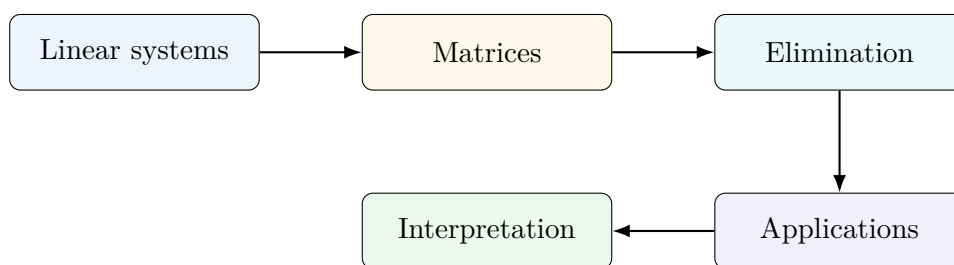


Chapter 1: Systems of Linear Equations

Sections 1.1–1.3

Theory, Visual Interpretation, Step-by-Step Computation, and Applications



MATH 3260 – Linear Algebra

Chapter Notes

Original examples with occasional Spurs-inspired values and standard applications.

Contents

1	Section 1.1: Systems of Linear Equations	2
1.1	Linear equations, systems, and solution sets	2
1.2	A unique solution	3
1.3	Geometric interpretation in two variables	3
1.4	Parametric descriptions in three variables	4
1.5	Original application examples	4
1.6	A detailed preview of elimination	5
1.7	Terminology summary	5
2	Section 1.2: Gaussian Elimination and Gauss-Jordan Elimination	6
2.1	Matrices as an organizing language	6
2.2	Elementary row operations	7
2.3	Row-echelon and reduced row-echelon form	7
2.4	Gaussian elimination: a Spurs-inspired three-item bundle	7
2.5	Gauss-Jordan elimination: a three-variable system	8
2.6	Inconsistent, dependent, and homogeneous systems	8
2.7	Terminology summary	9
3	Section 1.3: Applications of Linear Systems	11
3.1	A modeling workflow	11
3.2	Linear interpolation through two points	11
3.3	Quadratic interpolation through three points	12
3.4	Traffic-network flow	13
3.5	Kirchhoff's laws: an electrical circuit	14
3.6	Color-sensor calibration	14
3.7	Resource-allocation consistency check	14
3.8	Terminology summary	15
4	Chapter 1 closing summary	15

1 Section 1.1: Systems of Linear Equations

Learning objectives

By the end of this section, students should be able to:

1. recognize linear and nonlinear equations;
2. explain what it means to solve a system;
3. distinguish one solution, no solution, and infinitely many solutions;
4. connect algebraic solutions with intersections of lines and planes;
5. write parametric solution sets;
6. classify systems as consistent or inconsistent;
7. perform an introductory elimination calculation and verify a solution.

1.1 Linear equations, systems, and solution sets

Linear equation

A linear equation in variables $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where the coefficients $a_1, \dots, a_n$ and the constant b are fixed real numbers. Variables occur only to the first power and are not multiplied together.

System and solution

A **system of linear equations** is a collection of linear equations involving the same variables. A **solution** is an ordered list of values that makes every equation true simultaneously. The **solution set** is the collection of all solutions.

Linear or nonlinear?

Equation	Classification
$4x - 3y = 21$	Linear
$x^2 + y = 9$	Nonlinear: a variable is squared
$2x_1 - x_2 + x_3 = 20$	Linear
$xy - z = 1$	Nonlinear: variables are multiplied
$5u - 2v + 7w = 0$	Linear and homogeneous
$\sin x + y = 3$	Nonlinear: a trigonometric term appears

1.2 A unique solution

One solution

Solve

$$\begin{cases} u + v = 21, \\ u - v = -3. \end{cases}$$

Add the equations:

$$(u + v) + (u - v) = 21 + (-3),$$

so $2u = 18$ and $u = 9$. Substituting into $u + v = 21$ gives $v = 12$. Therefore,

$$(u, v) = (9, 12).$$

The values satisfy both equations. The numbers are Spurs-inspired, but the system is an original teaching example.

1.3 Geometric interpretation in two variables

Each linear equation in two variables represents a line. The solution set of a system is the set of intersection points shared by all lines.

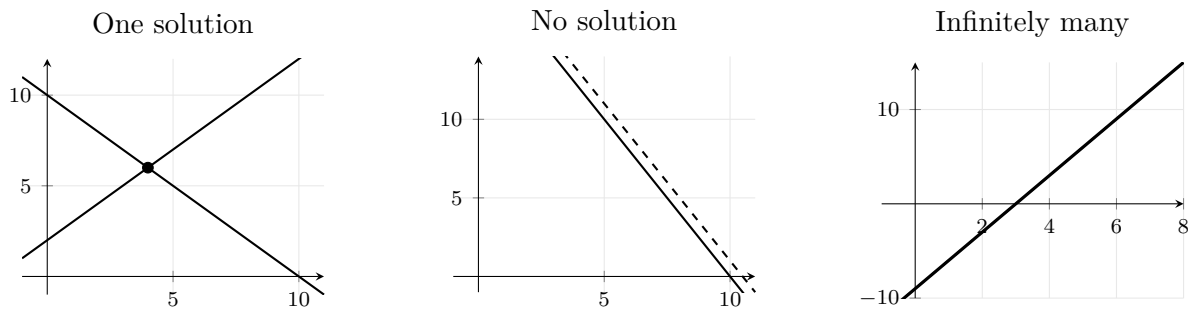


Figure 1: The three possible outcomes for two equations in two variables.

Consistency

A system is **consistent** if it has at least one solution. It is **inconsistent** if it has no solution.

No solution

Consider

$$\begin{cases} 2x + y = 20, \\ 4x + 2y = 42. \end{cases}$$

Twice the first equation gives $4x + 2y = 40$, but the second requires the same expression to equal 42. The system is inconsistent.

Infinitely many solutions

Consider

$$\begin{cases} 3x - y = 9, \\ 6x - 2y = 18. \end{cases}$$

The second equation is twice the first. Let $x = t$. Then $y = 3t - 9$, so

$$(x, y) = (t, 3t - 9), \quad t \in \mathbb{R}.$$

1.4 Parametric descriptions in three variables

Consider one equation in three variables:

$$x + 2y - z = 21.$$

Let $y = s$ and $z = t$. Then $x = 21 - 2s + t$, so

$$(x, y, z) = (21 - 2s + t, s, t), \quad s, t \in \mathbb{R}.$$

Geometrically, the solution set is a plane in $\mathbb{R}^3$.

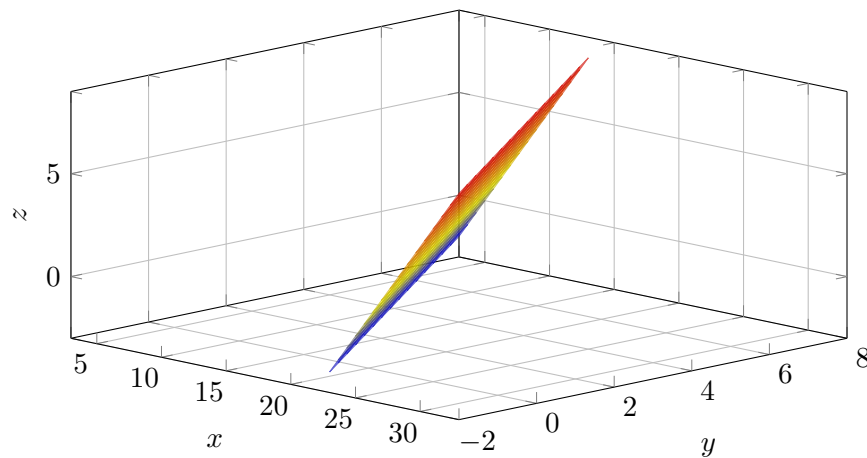


Figure 2: A portion of the plane $x + 2y - z = 21$.

1.5 Original application examples

Spurs-inspired fan-shop bundle

Let c be a modeled cap price and p a modeled poster price. Suppose

$$\begin{cases} 2c + p = 30, \\ c + 2p = 33. \end{cases}$$

Doubling the second equation and subtracting the first gives $3p = 36$, so $p = 12$ and $c = 9$. These are fictional teaching prices.

Two-category planning model

Let x and y be two categories in a planning model. Suppose the total count is 21, and a weighted count assigns weight 2 to the second category:

$$\begin{cases} x + y = 21, \\ x + 2y = 30. \end{cases}$$

Subtracting gives $y = 9$, and then $x = 12$. This is a generic teaching model.

1.6 A detailed preview of elimination

Solve

$$\begin{cases} x + y + z = 31, \\ 2x - y + 3z = 36, \\ -x + 2y + z = -2. \end{cases}$$

Start with

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 31 \\ 2 & -1 & 3 & 36 \\ -1 & 2 & 1 & -2 \end{array} \right].$$

Use $R_2 \leftarrow R_2 - 2R_1$:

$$[2, -1, 3 \mid 36] - 2[1, 1, 1 \mid 31] = [0, -3, 1 \mid -26].$$

Use $R_3 \leftarrow R_3 + R_1$:

$$[-1, 2, 1 \mid -2] + [1, 1, 1 \mid 31] = [0, 3, 2 \mid 29].$$

Then $R_3 \leftarrow R_3 + R_2$:

$$[0, 3, 2 \mid 29] + [0, -3, 1 \mid -26] = [0, 0, 3 \mid 3].$$

Thus $z = 1$. From $-3y + z = -26$, $y = 9$. From $x + y + z = 31$, $x = 21$. Therefore,

$$(x, y, z) = (21, 9, 1).$$

Common mistakes

- Checking a candidate in only one equation.
- Assuming every system has exactly one solution.
- Giving one sample point when a parameterized family is required.
- Confusing “many equations” with “independent information.”
- Ignoring the real-world meaning or units of a modeled solution.

1.7 Terminology summary

Term	Meaning
Linear equation	Equation of the form $a_1x_1 + \cdots + a_nx_n = b$.
System of linear equations	Several linear equations involving the same variables.
Solution	Values satisfying every equation simultaneously.
Solution set	Collection of all solutions.
Consistent system	System with at least one solution.
Inconsistent system	System with no solution.
Parameter	A variable used to describe a family of solutions.
Parametric form	Description of a solution set using one or more parameters.
Homogeneous equation	Linear equation with zero on the right-hand side.

2 Section 1.2: Gaussian Elimination and Gauss-Jordan Elimination

Learning objectives

Students should be able to:

1. identify matrix dimensions and entries;
2. construct coefficient and augmented matrices;
3. perform all three elementary row operations;
4. distinguish REF from RREF;
5. solve systems using Gaussian elimination and back-substitution;
6. solve systems using Gauss-Jordan elimination;
7. recognize inconsistent, dependent, and homogeneous systems.

2.1 Matrices as an organizing language

Matrix terminology

An $m \times n$ matrix has m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

The entry a_{ij} lies in row i and column j .

$$\begin{array}{c} 2 \text{ rows} \left\{ \begin{bmatrix} 21 & 9 & 1 \\ 20 & 12 & 6 \end{bmatrix} \right. \\ \underbrace{\hspace{10em}}_{3 \text{ columns}} \end{array} \quad \leftarrow a_{23} = 6$$

Figure 3: A 2×3 matrix and one indexed entry.

For

$$\begin{cases} x + 2y - z = 21, \\ 3x - y + 4z = 20, \end{cases}$$

the coefficient matrix and augmented matrix are

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{bmatrix}, \quad [A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 2 & -1 & 21 \\ 3 & -1 & 4 & 20 \end{array} \right].$$

2.2 Elementary row operations

The three legal operations

1. Row replacement: $R_i \leftarrow R_i + kR_j$.
2. Row interchange: $R_i \leftrightarrow R_j$.
3. Row scaling: $R_i \leftarrow kR_i$, where $k \neq 0$.

Each operation preserves the solution set of the associated linear system.

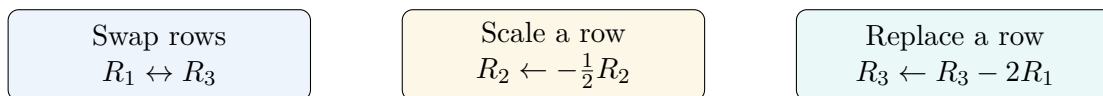


Figure 4: Elementary row operations.

2.3 Row-echelon and reduced row-echelon form

REF

A matrix is in **row-echelon form** if:

1. all nonzero rows lie above zero rows;
2. each leading entry is to the right of the leading entry in the row above;
3. all entries below each leading entry are zero.

RREF

A matrix is in **reduced row-echelon form** if it is in REF and, additionally:

1. each leading entry equals 1;
2. each leading 1 is the only nonzero entry in its column.

2.4 Gaussian elimination: a Spurs-inspired three-item bundle

Fictional fan-shop bundle

Let c be a cap price, s a shirt price, and p a poster price. Suppose

$$\begin{cases} c + s + p = 50, \\ 2c + s + 3p = 101, \\ c + 2s + 4p = 133. \end{cases}$$

These are fictional teaching prices.

Start with

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 2 & 1 & 3 & 101 \\ 1 & 2 & 4 & 133 \end{array} \right].$$

Use $R_2 \leftarrow R_2 - 2R_1$:

$$[2, 1, 3 \mid 101] - 2[1, 1, 1 \mid 50] = [0, -1, 1 \mid 1].$$

Use $R_3 \leftarrow R_3 - R_1$:

$$[1, 2, 4 \mid 133] - [1, 1, 1 \mid 50] = [0, 1, 3 \mid 83].$$

Then $R_3 \leftarrow R_3 + R_2$:

$$[0, 1, 3 \mid 83] + [0, -1, 1 \mid 1] = [0, 0, 4 \mid 84].$$

Thus the row-echelon form is

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 4 & 84 \end{array} \right].$$

Back-substitution gives $p = 21$, then $s = 20$, and finally $c = 9$:

$$(c, s, p) = (9, 20, 21).$$

2.5 Gauss-Jordan elimination: a three-variable system

Consider the following three-variable system:

$$\begin{cases} x + y + z = 50, \\ x + 2y + z = 71, \\ 2x + y + 3z = 88. \end{cases}$$

Start with

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 1 & 2 & 1 & 71 \\ 2 & 1 & 3 & 88 \end{array} \right].$$

Forward elimination:

$$R_2 \leftarrow R_2 - R_1, \quad R_3 \leftarrow R_3 - 2R_1,$$

gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & 0 & 21 \\ 0 & -1 & 1 & -12 \end{array} \right].$$

Then $R_3 \leftarrow R_3 + R_2$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 50 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & 9 \end{array} \right].$$

Backward elimination:

$$R_1 \leftarrow R_1 - R_3, \quad R_1 \leftarrow R_1 - R_2,$$

produces

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 20 \\ 0 & 1 & 0 & 21 \\ 0 & 0 & 1 & 9 \end{array} \right].$$

Therefore,

$$(x, y, z) = (20, 21, 9).$$

2.6 Inconsistent, dependent, and homogeneous systems

Inconsistent row

For

$$\begin{cases} x + 2y = 21, \\ 2x + 4y = 43, \end{cases}$$

$R_2 \leftarrow R_2 - 2R_1$ creates the row

$$[0 \ 0 \mid 1],$$

which means $0 = 1$. Therefore the system is inconsistent.

Infinitely many solutions from RREF

For

$$\begin{cases} x + y + z = 21, \\ 2x + 2y + 2z = 42, \\ x - y + z = 1, \end{cases}$$

the RREF equations are $x + z = 11$ and $y = 10$. Let $z = t$:

$$(x, y, z) = (11 - t, 10, t), \quad t \in \mathbb{R}.$$

Homogeneous system

A system $A\mathbf{x} = \mathbf{0}$ is homogeneous. It always has the trivial solution $\mathbf{x} = \mathbf{0}$ and may also have nontrivial solutions.

For

$$\begin{cases} x + y - z = 0, \\ 2x + 2y - 2z = 0, \\ x - y + z = 0, \end{cases}$$

the RREF equations are $x = 0$ and $y - z = 0$. Let $z = t$:

$$(x, y, z) = (0, t, t) = t(0, 1, 1).$$

Common mistakes

- Applying a row operation to only part of a row.
- Forgetting the augmented column.
- Scaling a row by zero.
- Confusing REF with RREF.
- Interpreting $[0 \ 0 \ 0 \mid 0]$ as a contradiction.
- Setting a free variable to zero without explaining the full solution family.

2.7 Terminology summary

Term	Meaning
Matrix	Rectangular array of entries.
$m \times n$ matrix	Matrix with m rows and n columns.

Entry a_{ij}	Entry in row i , column j .
Coefficient matrix	Matrix containing only variable coefficients.
Augmented matrix	Coefficient matrix with the constants appended.
Elementary row operation	Row replacement, row interchange, or nonzero row scaling.
Row equivalent	Matrices connected by elementary row operations.
Pivot position	Location of a leading entry in echelon form.
REF	Row-echelon form.
RREF	Reduced row-echelon form.
Gaussian elimination	Row reduction to REF followed by back-substitution.
Gauss-Jordan elimination	Row reduction all the way to RREF.
Free variable	Variable corresponding to a nonpivot column.
Homogeneous system	System with zero right-hand side.
Trivial solution	All variables equal zero.

3 Section 1.3: Applications of Linear Systems

Learning objectives

Students should be able to:

1. translate an applied situation into variables, equations, and a matrix model;
2. fit lines and quadratic polynomials through data points;
3. use conservation laws in traffic and network models;
4. solve a simple circuit model using Kirchhoff's laws;
5. interpret parameters, units, and feasibility restrictions;
6. verify a model computationally without replacing the mathematical setup.

3.1 A modeling workflow

Five modeling steps

1. Define variables and units.
2. Translate each condition into a linear equation.
3. Assemble $A\mathbf{x} = \mathbf{b}$.
4. Solve or parameterize the system.
5. Check both mathematical correctness and real-world feasibility.

3.2 Linear interpolation through two points

Fit a line $y = mx + b$ through the original points

$$(1, 21), \quad (3, 9).$$

Substitution gives

$$\begin{cases} m + b = 21, \\ 3m + b = 9. \end{cases}$$

Subtracting gives $2m = -12$, so $m = -6$. Then $b = 27$:

$$y = -6x + 27.$$

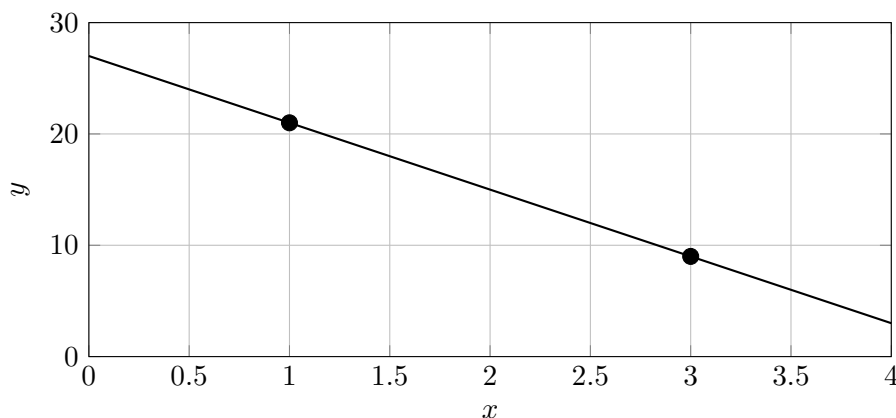


Figure 5: Exact linear interpolation through two points.

3.3 Quadratic interpolation through three points

Fit

$$p(x) = ax^2 + bx + c$$

through

$$(0, 9), \quad (1, 20), \quad (2, 21).$$

The system is

$$\begin{cases} c = 9, \\ a + b + c = 20, \\ 4a + 2b + c = 21. \end{cases}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 9 \\ 1 & 1 & 1 & 20 \\ 4 & 2 & 1 & 21 \end{array} \right].$$

Use $R_2 \leftarrow R_2 - R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 0 & 0 & 1 & 9 \\ 1 & 1 & 0 & 11 \\ 4 & 2 & 0 & 12 \end{array} \right].$$

Then $R_3 \leftarrow R_3 - 4R_2$:

$$[4, 2, 0 \mid 12] - 4[1, 1, 0 \mid 11] = [0, -2, 0 \mid -32].$$

Thus $b = 16$, $a = -5$, and $c = 9$:

$$\boxed{p(x) = -5x^2 + 16x + 9}.$$

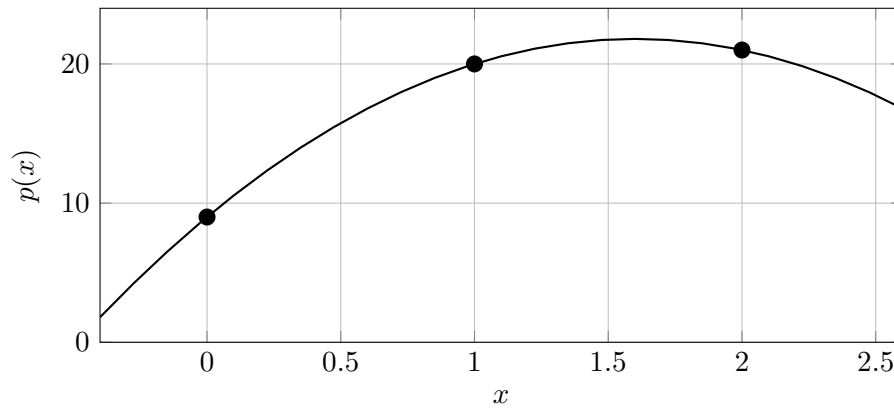


Figure 6: Quadratic interpolation through three data points.

3.4 Traffic-network flow

At every node,

$$\text{total inflow} = \text{total outflow}.$$

Consider the fictional traffic network:

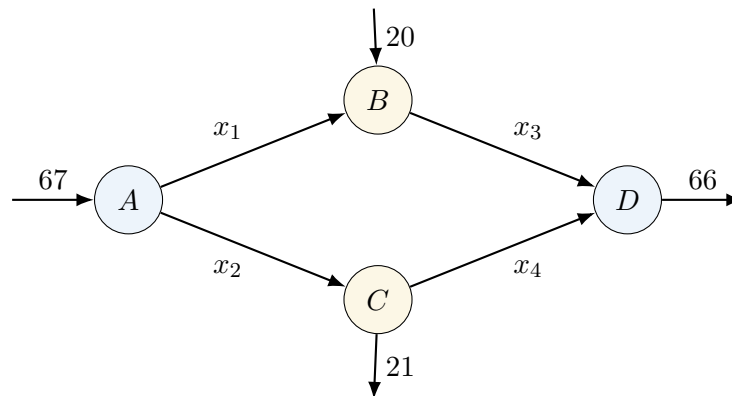


Figure 7: A fictional transportation network measured in vehicles per hour.

The node equations are

$$x_1 + x_2 = 67, \quad x_1 + 20 = x_3, \quad x_2 = x_4 + 21.$$

The fourth-node equation $x_3 + x_4 = 66$ is dependent on the first three. Let $x_1 = t$. Then

$$x_2 = 67 - t, \quad x_3 = t + 20, \quad x_4 = 46 - t.$$

Therefore,

$$(x_1, x_2, x_3, x_4) = (t, 67 - t, t + 20, 46 - t).$$

Nonnegative flows require

$$0 \leq t \leq 46.$$

3.5 Kirchhoff's laws: an electrical circuit

Circuit laws

- Kirchhoff's Current Law: current entering a junction equals current leaving it.
- Kirchhoff's Voltage Law: signed voltage changes around a closed loop sum to zero.
- Ohm's law: $V = RI$.

A simplified electrical-circuit model produces

$$\begin{cases} I_1 - I_2 + I_3 = 0, \\ 2I_1 + I_2 = 38, \\ I_2 + 3I_3 = 53. \end{cases}$$

Start with

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 38 \\ 0 & 1 & 3 & 53 \end{array} \right].$$

Use $R_2 \leftarrow R_2 - 2R_1$:

$$[2, 1, 0 \mid 38] - 2[1, -1, 1 \mid 0] = [0, 3, -2 \mid 38].$$

Then use the fraction-free operation $R_3 \leftarrow 3R_3 - R_2$:

$$3R_3 = [0, 3, 9 \mid 159],$$

so

$$[0, 3, 9 \mid 159] - [0, 3, -2 \mid 38] = [0, 0, 11 \mid 121].$$

Therefore $I_3 = 11$. From $3I_2 - 2I_3 = 38$, $I_2 = 20$. From $I_1 - I_2 + I_3 = 0$, $I_1 = 9$:

$$(I_1, I_2, I_3) = (9, 20, 11).$$

3.6 Color-sensor calibration

Suppose a sensor has unknown red, green, and blue response weights (r, g, b) and three calibration measurements produce

$$\begin{cases} r + g + b = 30, \\ 2r + g + 3b = 41, \\ r + 4g + 2b = 91. \end{cases}$$

This model can be solved by row reduction or verified computationally. The essential modeling step is recognizing that each calibration measurement is linear in the unknown weights.

3.7 Resource-allocation consistency check

A resource-allocation model uses variables u, v, w and reports:

$$\begin{cases} u + v + w = 9, \\ 2u + v = 12, \\ v + 2w = 8. \end{cases}$$

Substituting $v = 12 - 2u$ and $w = u - 2$ into the first equation gives

$$u + (12 - 2u) + (u - 2) = 9,$$

which simplifies to the contradiction $10 = 9$. Therefore, the three planning statements are inconsistent. This is a useful model-checking example: at least one reported total must be corrected before a feasible allocation can be found.

Modeling cautions

- Define variables and units before writing equations.
- Distinguish exact interpolation from approximate regression.
- Check whether parameter values make flows nonnegative.
- A mathematically correct solution can be physically impossible.
- Verify dependent equations rather than treating them as new independent information.

3.8 Terminology summary

Term	Meaning
Mathematical model	Mathematical representation of a real or designed situation.
Interpolation	Construction of a function passing exactly through given data points.
Polynomial coefficients	Unknown constants multiplying the powers of the variable.
Conservation law	Principle that total inflow equals total outflow at a node.
Network flow	Quantities moving through connected nodes and directed edges.
Dependent equation	Equation that adds no new independent constraint.
Feasible solution	Mathematical solution satisfying contextual restrictions.
Kirchhoff's Current Law	Current entering a junction equals current leaving.
Kirchhoff's Voltage Law	Signed voltage changes around a loop sum to zero.
Calibration	Estimation of model parameters from known inputs and measured outputs.

4 Chapter 1 closing summary

The chapter storyline

1. Linear systems encode simultaneous constraints.
2. Geometry explains one, none, or infinitely many solutions.
3. Matrices organize coefficients and constants.
4. Elementary row operations preserve the solution set.
5. Gaussian elimination reaches REF; Gauss-Jordan reaches RREF.
6. Free variables create parametric families.
7. Applications require variable definitions, equations, computation, and interpretation.
8. Python is most useful when it verifies, visualizes, or explores mathematics that has already been modeled carefully.